Income, Child Mortality and Life Expectancy

A CRISP-DM walkthrough, from the question to the deployed app

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The question

Do richer countries really have fewer children dying and people living longer, and has child mortality dropped since 2000?

If income and health move together, help goes to the poorest countries first. If not, funders should back immunisation and basic healthcare.

MY HUNCH (BEFORE THE DATA)

- Richer countries have far lower child mortality
- They also live longer
- Child mortality has fallen in most countries since 2000

Four World Bank series, one skewed outcome

- Life expectancy, under-5 mortality, GDP per capita (current US$), population
- 262 countries and aggregates × 66 years (1960 to 2025), loaded and profiled one source at a time
- Missing values are heaviest in the early years
- A few very high mortality values are genuine country-years, so I kept them

68.6 vs 37.4

mean vs median under-5 mortality: strongly right-skewed, so I use medians and log mortality

Cleaning and joining: 17,292 rows down to 10,525

17,292
rows per source (262 entities × 66 years)

›

17,292
after the inner join on country code + year: none lost or duplicated

›

12,065
rows with all four indicators present

›

10,525
after removing aggregates: 194 countries

- Missing markers became real missing values; each source was reshaped from wide to long
- Incomplete rows were dropped, not filled: filling would flatten the relationships I am testing
- Added log GDP and GDP per capita quartiles (four equal-sized income groups)

Four hypothesis tests, matched to the question

Question	Test	Why this test
Does child mortality differ across four income groups?	One-way ANOVA (log mortality)	Numeric outcome, four groups
Did mortality fall between 2000 and 2024?	Paired t-test	Same countries measured twice
Is income group linked to life expectancy of 70+?	Chi-square (2024)	Two categorical variables
Do richer and poorer halves differ in meeting the UN target?	Two-proportion Z-test (2024)	A/B-style comparison of two groups

α = 0.05. H0 and H1 were written before each test was run.

ANOVA: child mortality across income groups

HYPOTHESES (BEFORE RUNNING)

H0: mean log child mortality is the same in all four GDP quartiles.

H1: at least one quartile has a different mean.

One-way ANOVA on log mortality: numeric outcome, four groups. All 10,525 rows.

RESULT (α = 0.05)

$\eta^2 = 0.73$

F(3, 10521) = 9,557.7
p < 0.001
Median per 1,000: 158.1, 59.2, 22.6, 8.0 (Q1 to Q4)

Reject H0

WHAT THE P-VALUE DOES NOT MEAN

It is not the probability that H0 is true or that the result is due to chance, and it does not give the size: that comes from η^2 and the medians. Countries repeat across years, so it looks more certain than it is. It is not proof of causation.

Paired t-test: child mortality, 2000 vs 2024

HYPOTHESES (BEFORE RUNNING)

H0: the mean change in log child mortality between 2000 and 2024 is zero.

H1: the mean change is not zero.

Paired t-test on log mortality: the same 186 countries measured twice.

RESULT (α = 0.05)

d = 1.75

t(185) = 23.8
p = 2.64e-58
97% of countries fell; 2024 is about 46% of 2000

Reject H0

WHAT THE P-VALUE DOES NOT MEAN

It is not the probability that H0 is true or that the fall happened by chance. It says nothing about why mortality fell, and the size comes from d and the 97%, not from the p-value. Only countries with data in both years are included.

Chi-square: income group and life expectancy of 70+

HYPOTHESES (BEFORE RUNNING)

H0: GDP quartile and life expectancy category (70+ vs under 70) are independent.

H1: they are associated.

Chi-square test of independence, 2024 only: 188 countries, each counted once.

RESULT (α = 0.05)

V = 0.72

$\chi^2(3) = 97.1$
p = 6.39e-21
Reach 70: 13%, 68%, 89%, 100% (Q1 to Q4)

Reject H0

WHAT THE P-VALUE DOES NOT MEAN

It is not the probability that H0 is true or that the association is due to chance, and it does not say how strong it is: Cramér's V and the shares do. It does not say which quartiles differ or that income causes longer life. The 70-year cutoff loses detail.

A/B-style Z-test: richer vs poorer half

HYPOTHESES (BEFORE RUNNING)

H0: the share of countries meeting the UN target is the same in both halves.

H1: the shares differ.

Two-proportion Z-test, 2024: below vs above median GDP, 94 countries each. Target: 25 or fewer under-5 deaths per 1,000.

RESULT (α = 0.05)

54 pts

39% vs 94% meet the target (95% CI 43 to 65)
z = 7.88, p = 3.27e-15
Fisher check: p = 4.59e-16

Reject H0

WHAT THE P-VALUE DOES NOT MEAN

It is not the probability that H0 is true or that the gap is due to chance, and it does not give the gap's size. Groups were not randomly assigned, so a significant result shows the groups differ. It does not by itself prove that income caused the difference, the way a randomized A/B test could.

All four tests reject H0

ANOVA: CHILD MORTALITY

$\eta^2 = 0.73$

$F(3, 10521) = 9{,}557.7$, $p < 0.001$. Median 158 → 8 per 1,000 from lowest to highest quartile.

PAIRED T-TEST: 2000 vs 2024

97%

of countries fell; 2024 is about 46% of 2000. $t(185) = 23.8$, $p = 2.64e\text{-}58$, $d = 1.75$.

CHI-SQUARE: LIFE EXPECTANCY 70+

V = 0.72

13% of lowest-quartile countries reach 70, vs 100% of the highest. $\chi^2(3) = 97.1$, $p = 6.39e\text{-}21$.

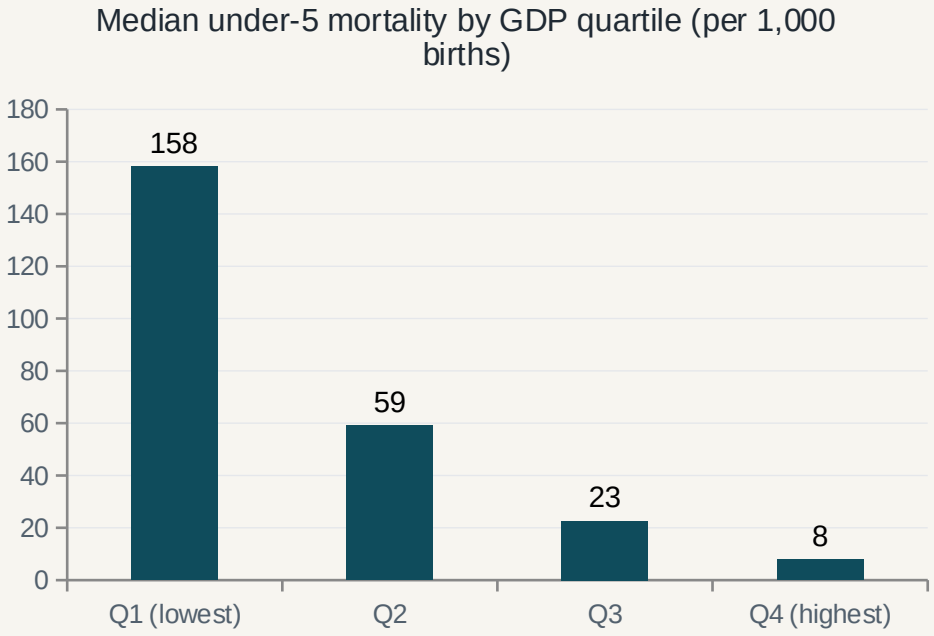
A/B Z-TEST: RICHER vs POORER HALF

54 pts

39% vs 94% meet the target (95% CI 43 to 65). $z = 7.88$, $p = 3.27e\text{-}15$.

A p-value is not the probability that H0 is true, and it does not measure effect size or prove causation.

What it means, and what to do



RECOMMENDATION

- Prioritise the lowest-income quartile: median 158 vs 8 deaths per 1,000
- Do not rely on growth alone: mortality fell in 97% of countries, yet the poorest still lag, which supports health programmes like immunisation and primary care
- Caution: these are associations, and the groups were not randomly assigned

Thank you

Questions & answers

- The data and how I cleaned and joined it
- Why these four tests
- What the results do and do not show

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